

8257 Direct Memory Access Controller (DMAC) -- MPCA Module 4

Definition: The Intel 8257 is a 4-channel programmable Direct Memory Access Controller (DMAC) that enables high-speed data transfer directly between I/O peripherals and system RAM without CPU intervention, bypassing the slow CPU fetch-execute cycle.

1. Detailed Technical Explanation

1. The DMA Concept & Speed Advantage

In standard programmed I/O, every byte transferred from a disk/network card to RAM requires multiple CPU instructions (IN AL, DX followed by MOV [SI], AL). DMA bypasses the CPU:

PROGRAMMED I/O:	Peripheral ---> CPU (AL Reg) ---> RAM	(Slow: ~20-50 Clock cycles/byte)
DMA TRANSFER:	Peripheral =====> RAM	(Fast: 2-4 Clock cycles/byte)

2. 8257 Internal Block Diagram & Channels

8257 DMAC ARCHITECTURE

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| [ Channel 0 ] -> 16-bit DMA Address Reg & 14-bit Count Reg |
| [ Channel 1 ] -> 16-bit DMA Address Reg & 14-bit Count Reg |
| [ Channel 2 ] -> 16-bit DMA Address Reg & 14-bit Count Reg |
| [ Channel 3 ] -> 16-bit DMA Address Reg & 14-bit Count Reg |
|                                                              |
| Control Logic & Priority Resolver:                          |
|   - DRQ0 to DRQ3 (DMA Requests from Peripherals)          |
|   - /DACK0 to /DACK3 (DMA Acknowledgements to Peripherals)|
|   - HRQ (Hold Request to 8086 CPU)                         |
|   - HLDA (Hold Acknowledge from 8086 CPU)                  |
|   - /MEMR, /MEMW, /IOR, /IOW (Bus Control Strokes)         |
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3. DMA Transfer Sequence (Step-by-Step)

1. Peripheral asserts DMA Request (DRQ) to 8257.

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4. DMA Transfer Modes

1. Single Transfer Mode: 8257 transfers one byte per request, releasing the bus back to CPU between transfers.

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5. Must-Write Points for Exams

- * 8257 has 4 independent DMA channels (Channel 0 to Channel 3).

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6. Quick Recall Flow

DRQ -> HRQ -> CPU floats bus & asserts HLDA -> DACK -> Single Cycle MEM/IO Read/Write -> Terminal Count (TC)